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Normalization:
Normalization is the process of efficiently
organizing data in a database.
There are two goals of the normalization process are,
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- ☐ Eliminating redundant data (for example, storing the same data in more than one table) and
- ☐ Ensuring data dependencies make sense (only storing related data in a table).

Both of these are worthy goals as they reduce the amount of space a database consumes and ensure that data is logically stored.

There are several benefits for using Normalization in Database.

Benefits:

- a. Eliminate data redundancy
- b. Improve performance
- c. Query optimization
- d. Faster update due to less number of columns in one table
- e. Index improvement

There are different types of Normalizations form available in the Database. Let's see one by one.

1. First Normal Form (1NF):
First normal form (1NF) sets the very basic rules for an organized database:

- Eliminate duplicative columns from the same table.
- Create separate tables for each group of related data and identify each row with a unique column or set of columns (the primary key).
 - a. Remove repetitive groups
 - b. Create Primary Key

Example:

Name	State	Country	Phone1	Phone2	Phone3
John	101	1	488-511-3258	781-896-9897	425-983-9812
Bob	102	1	861-856-6987		
Rob	201	2	587-963-8425	425-698-9684	
PK	[Phone No's]				
?	?				
ID	Name	State	Country	Phone	
1	John	101	1	488-511-3258	
2	John	101	1	781-896-9897	
3	John	101	1	425-983-9812	
4	Bob	102	1	861-856-6987	
5	Rob	201	2	587-963-8425	
6	Rob	201	2	425-698-9684	

2. Second Normal Form (2NF):
Second normal form (2NF) further addresses the concept of removing duplicative data:

- Meet all the requirements of the first normal form.
- Remove subsets of data that apply to multiple rows of a table and place them in separate tables.

Remove columns which create duplicate data in a table and related a new table with Primary Key - Foreign Key relationship

ID	Name	State	Country	Phone
1	John	101	1	488-511-3258
2	John	101	1	781-896-9897
3	John	101	1	425-983-9812
4	Bob	102	1	861-856-6987
5	Rob	201	2	587-963-8425
6	Rob	201	2	425-698-9684

3. Third Normal Form (3NF):

Third normal form (3NF) goes one large step further:

- Meet all the requirements of the second normal form.
- Remove columns those are not dependent upon the primary key. Country can be derived from State also... so removing country

ID	Name	State	Country
1	John	101	1
2	Bob	102	1
3	Rob	201	2

4. Fourth Normal Form (4NF):

Finally, fourth normal form (4NF) has one additional requirement:

- Meet all the requirements of the third normal form.
- A relation is in 4NF if it has no multi-valued dependencies.

If PK is composed of multiple columns then all non-key attributes should be derived from FULL PK only.

If some non-key attribute can be derived from partial PK then remove it. The 4NF also known as BCNF NF

ID	Name	State
1	John	101
2	Bob	102
3	Rob	201

5. Fifth Normal Form (5NF):

A database table is said to be in 5NF if it is in 4NF and contains no redundant values or

We can also said a table to be in 5NF if it is in 4NF and contains no join dependencies.

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